

Documentation Standards

Field	Value
Revision	1
Created	2026-06-07
Last modified	2026-06-07
Status	active
Status summary	Initial documentation standards for the Helix OTA corpus. Defines mandatory metadata, ToC, file naming, multi-format export targets, diagram conventions, cross-reference style, and the anti-bluff/UNVERIFIED convention.
Issues	None known at Revision 1.
Fixed	N/A (initial revision).
Continuation	Validate export tooling (renderable vs source-only matrix) against an actual build; reconcile §11.4.x clause numbers against the authoritative HelixConstitution once available (UNVERIFIED).

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The table-of-contents requirement is mandated by HelixConstitution §11.4.61. Every corpus document MUST carry a ToC immediately after its metadata table.

1. Purpose and scope

This document defines the documentation standards for the Helix OTA documentation corpus (rooted at docs/research/main_specs/). It is normative: every document authored or revised in the corpus MUST conform to the rules below. Where a rule references a HelixConstitution clause (e.g. §11.4.29, §11.4.61, §7.1, §11.4.6, §11.4.123), that clause is the controlling authority and this document is its corpus-local restatement.

The exact wording and numbering of the cited HelixConstitution clauses has not been cross-checked against the authoritative constitution text in this revision and is marked UNVERIFIED (see §8).

2. Mandatory metadata table

Every document MUST begin with a metadata table as its first content element, before any prose, the ToC, or any heading other than the document title. The table MUST contain, at minimum, the following rows in this order:

Field	Meaning
Revision	Monotonically increasing integer, starting at 1. Bump on every substantive change.
Created	ISO-8601 date (YYYY-MM-DD) the document was first created. Never changes.
Last modified	ISO-8601 date (YYYY-MM-DD) of the most recent substantive change.
Status	One of: draft, active, deprecated, superseded.
Status summary	One- to three-sentence description of the document's current state and intent.
Issues	Known problems, gaps, or open questions. Use None known at Revision N. when empty.
Fixed	What was corrected since the previous revision. Use N/A (initial revision). at Revision 1.
Continuation	Concrete follow-up work required. Forward-looking TODOs live here, not in prose.

Rules:

- The metadata table format is fixed (two columns: Field, Value). Do not reorder or rename the mandatory rows.
- Additional rows MAY be appended after Continuation (for example Owner, Reviewers), but the mandatory rows MUST all be present.

- Created is immutable. Last modified and Revision MUST be updated together on every change.

3. Table of contents requirement

Per HelixConstitution §11.4.61, a Table of contents is mandatory.

- The ToC MUST appear immediately after the metadata table and before the first body section.
- ToC entries MUST be ordered, numbered, and link to in-document anchors.
- The ToC MUST be kept in sync with the document's headings; a heading without a ToC entry, or a ToC entry without a heading, is a conformance defect.
- Documents short enough to fit on a single screen still require a ToC; there is no length exemption.

4. File naming

Per HelixConstitution §11.4.29, file names MUST use lowercase_snake_case.

- Allowed characters: lowercase ASCII letters a–z, digits 0–9, and underscore _.
- Words are separated by single underscores. No spaces, hyphens, camelCase, or uppercase letters.
- The extension is .md for Markdown sources.
- Examples (conformant): documentation_standards.md, ota_update_flow.md, event_bus_contract.md.
- Examples (non-conformant): Documentation-Standards.md, otaUpdateFlow.md, event bus contract.md.
- Numeric ordering prefixes for directories (for example 00-master/) are an established corpus convention and are permitted at the directory level; file basenames themselves remain lowercase_snake_case.

5. Multi-format export targets

The corpus targets the following export and diagram formats. Each is classified as **renderable** (produced by an export/render step today) or **source-only** (authored artifact that is the source of truth but not auto-rendered in this revision).

Target	Category	Status	Notes
PDF	Document export	Renderable (UNVERIFIED)	Print/archival output derived from Markdown. Pipeline not verified in this revision.
HTML	Document export	Renderable (UNVERIFIED)	Web/browsable output derived from Markdown. Pipeline not

Target	Category	Status	Notes
DOCX	Document export	Renderable (UNVERIFIED)	verified in this revision. Word-compatible output for external distribution. Pipeline not verified in this revision.
Mermaid (.mmd)	Diagram source	Source-only (source of truth)	Authored, version-controlled diagram source. See §6.
draw.io (.drawio)	Diagram source	Source-only	Editable diagram source where Mermaid is insufficient.
SVG (.svg)	Diagram render	Renderable	Vector render, typically generated from Mermaid or draw.io.
UML (.uml/ PlantUML)	Diagram source	Source-only	Authored UML source.
PNG (.png)	Diagram render	Renderable	Raster render for embedding where vector is not supported.

Notes:

- The renderability of PDF/HTML/DOCX depends on the corpus build tooling. Until that tooling is exercised and confirmed, these three are marked **Renderable (UNVERIFIED)** per §8.
- Diagram **sources** (Mermaid, draw.io, UML) are never deleted in favor of their rendered outputs; the rendered SVG/PNG are derived artifacts.

6. Diagram conventions

- **Mermaid is the source of truth** for diagrams. Author diagrams in Mermaid wherever the diagram type is expressible in Mermaid (flowcharts, sequence, class, state, ER, etc.).
- SVG and PNG outputs are **generated** from the Mermaid source; do not hand-edit the rendered SVG/PNG. If a render is wrong, fix the Mermaid source and re-render.
- Use draw.io or UML/PlantUML sources only when Mermaid cannot express the required diagram; in that case the draw.io/UML file is the source of truth for that specific diagram and the same “edit source, not render” rule applies.

- Embed diagrams in Markdown either as fenced mermaid blocks (preferred, so the source travels with the prose) or by referencing the generated SVG/PNG with a path to the committed render.
- Diagram source files follow the §4 naming rules.

7. Cross-reference style

- **Intra-document** references use Markdown anchor links to numbered sections, e.g. [see §5](#5-multi-format-export-targets).
- **Inter-document** references within the corpus use relative paths from the referencing file, e.g. [OTA update flow](../additions/ota_update_flow.md), optionally with a section anchor.
- **Clause references** to the HelixConstitution use the § notation with the dotted clause number, e.g. §11.4.29, §11.4.61, §7.1. Always cite the clause when restating a normative rule it controls.
- Do not use bare URLs in prose; wrap them in descriptive Markdown link text.
- A reference whose target has not been confirmed to exist **MUST** be marked UNVERIFIED (§8) rather than presented as a confirmed link.

8. Anti-bluff and UNVERIFIED convention

Per HelixConstitution §7.1, §11.4.6, and §11.4.123, the corpus enforces an anti-bluff rule:

- **No fabricated facts.** Do not state numbers, behaviors, file paths, API signatures, or capabilities that have not been confirmed from a real source.
- **No fabricated citations.** Do not invent references, clause numbers, document titles, or external sources. If a citation cannot be confirmed, mark it UNVERIFIED.
- **Mark the unconfirmed.** Any claim, figure, citation, or cross-reference that has not been verified **MUST** be tagged inline with the literal token UNVERIFIED, e.g. “the export pipeline emits DOCX (UNVERIFIED)”.
- **Prefer omission or marking over guessing.** When in doubt, either leave the item out or mark it UNVERIFIED; never present a guess as fact.
- Open verification work belongs in the Continuation row of the metadata table (§2).

In this revision, the following are explicitly UNVERIFIED: the exact text/numbering of all cited HelixConstitution clauses, and the renderable status of the PDF/HTML/DOCX export pipelines (§5).

9. Submodule catalogue (canonical names)

Documents **MUST** refer to submodules using only the verified canonical names below. Do not invent, abbreviate, or rename submodules.

Core / backend submodules:

- auth
- security

- database
- Storage
- observability
- eventbus
- ratelimiter
- middleware
- http3
- mdns
- recovery
- Herald
- config
- discovery
- cache
- docs_chain / Document / Formatters
- containers

KMP (Kotlin Multiplatform) submodules:

- Auth-KMP
- Security-KMP
- Storage-KMP
- Config-KMP

Any submodule not on this list **MUST NOT** be referenced as if it exists; if one is needed, add it here first (and mark it **UNVERIFIED** until confirmed).

10. Authoring checklist

Before committing a corpus document, confirm:

☐

Metadata table is the first element and contains all mandatory rows (§2).

☐

Revision and Last modified were bumped together for this change (§2).

☐

Table of contents is present, numbered, and in sync with headings (§3, §11.4.61).

☐

File name is lowercase_snake_case.md (§4, §11.4.29).

☐

Diagrams are authored in Mermaid (or justified draw.io/UML) as source of truth; renders are generated, not hand-edited (§6).

☐

Cross-references use the correct intra/inter-document and § clause styles (§7).

☐

Every unconfirmed fact, figure, or citation is tagged **UNVERIFIED** (§8).

☐

Only canonical submodule names from §9 are used.